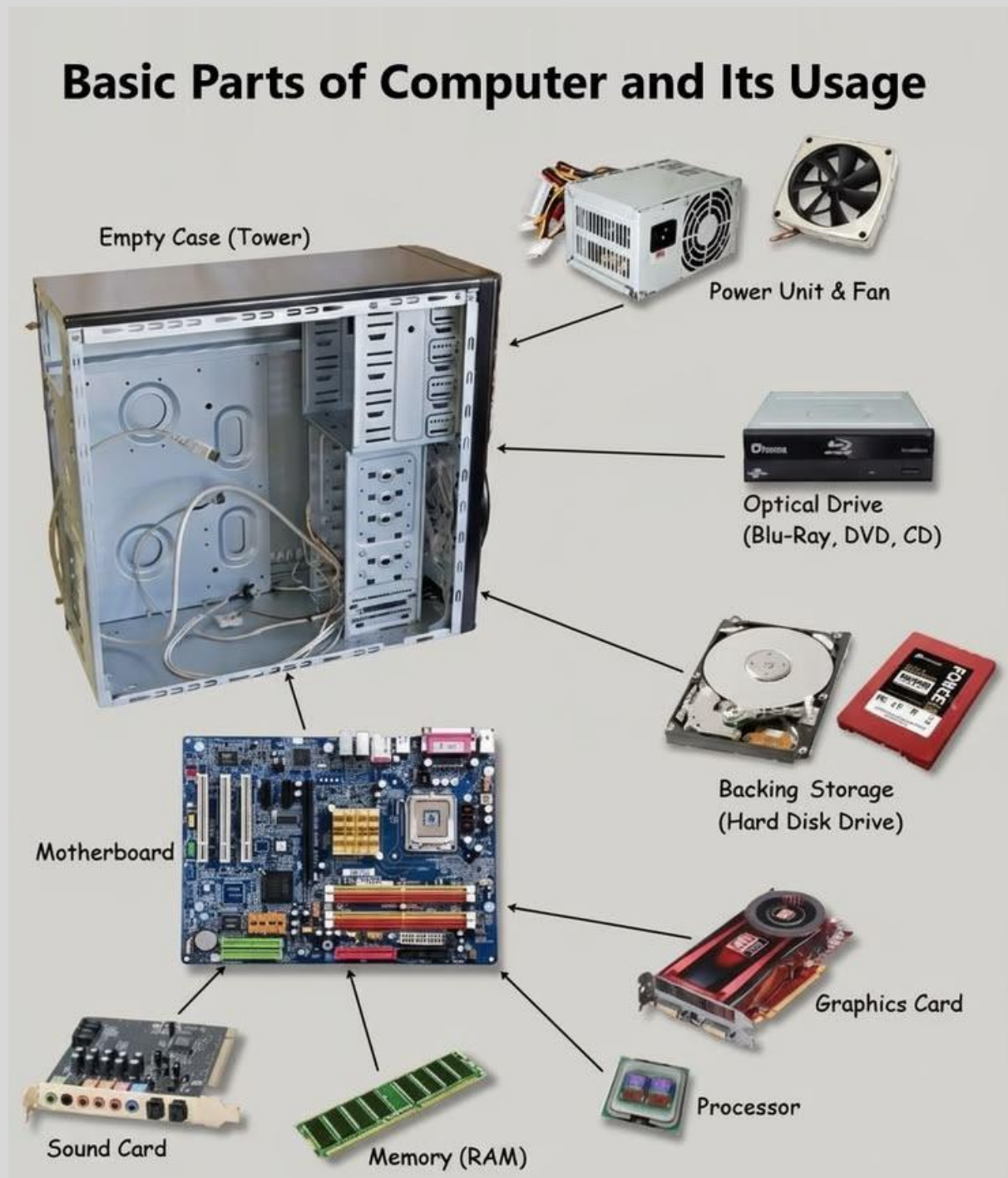


# HARDWARE

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



## 1. Empty Case (Tower)

- The **case (cabinet)** is the outer body of the computer.
  - It holds and protects all internal components.
  - It also helps with airflow and cooling.
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## 2. Power Unit & Fan (PSU)

- The **Power Supply Unit (PSU)** provides electricity to all parts of the computer.
  - It converts AC power into DC power.
  - The **fan** keeps the system cool and prevents overheating.
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## 3. Optical Drive (CD/DVD/Blu-ray)

- Used to **read and write discs** like CDs, DVDs, and Blu-rays.
  - Mostly used for installing software, movies, or data transfer.
  - Now less common in modern computers.
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## 4. Storage Devices (HDD & SSD)

- **HDD (Hard Disk Drive):**
    - Stores large amounts of data.
    - Uses spinning disks (mechanical).
  - **SSD (Solid State Drive):**
    - Faster storage with no moving parts.
    - Improves speed and performance.
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## 5. Motherboard

- The **main circuit board** of the computer.
  - Connects all components like CPU, RAM, storage, and GPU.
  - Allows communication between all parts.
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## 🔊 6. Sound Card

- Controls **audio input and output**.
  - Used for speakers, headphones, and microphones.
  - Many modern motherboards have built-in sound cards.
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## ⚡ 7. Memory (RAM)

- RAM stores **temporary data** while the computer is running.
  - Helps in fast performance and multitasking.
  - More RAM = smoother performance.
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## 🎮 8. Graphics Card (GPU)

- Handles **graphics and visuals**.
  - Important for gaming, video editing, and design.
  - Produces high-quality images and smooth gameplay.
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## 📦 9. Processor (CPU)

- The **brain of the computer**.
- Executes instructions and processes data.
- Controls all operations of the system.

2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.  
  
Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

### HDD vs SSD – Comparison Table

| Factor              | HDD (Hard Disk Drive)  | SSD (Solid State Drive)  |
|---------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Speed</b>        | Slower (uses spinning disks)                                                                            | Faster (uses flash memory, no moving parts)                                                                 |
| <b>Durability</b>   | Less durable (can get damaged easily if dropped)                                                        | More durable (shock-resistant, no moving parts)                                                             |
| <b>Storage Cost</b> | Cheaper, more storage for less price                                                                    | Expensive, less storage for same price                                                                      |
| <b>Noise</b>        | Makes noise while working                                                                               | Silent (no moving parts)                                                                                    |
| <b>Power Use</b>    | Uses more power                                                                                         | Uses less power                                                                                             |
| <b>Best Use</b>     | Storing large files (movies, backups)                                                                   | Fast performance (gaming, OS, software)                                                                     |

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

### CPU (Central Processing Unit)

The CPU acts as the brain of the computer, handling all game instructions such as player actions, enemy AI, and physics. In games like PUBG, it ensures smooth gameplay by processing everything happening in the background. A faster CPU helps reduce lag and improves overall game performance.

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### ⚡ RAM (Random Access Memory)

RAM stores temporary game data like maps, textures, and ongoing actions while the game is running. It allows the game to load quickly and run smoothly without interruptions. More RAM helps prevent lag and improves multitasking during gameplay.

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### 🎮 GPU (Graphics Processing Unit)

The GPU is responsible for rendering all visuals, including characters, environments, and effects. It ensures high-quality graphics and smooth frame rates in games like Free Fire. A powerful GPU gives better visuals and a more realistic gaming experience.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

#### 1. Graphics Card (GPU)

- The GPU is the **most important component for gaming**.
  - Upgrading it will improve **graphics quality, resolution, and frame rate (FPS)**.
  - Helps FIFA run smoothly with realistic visuals and no stuttering.
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#### 2. Processor (CPU)

- A better CPU improves **game logic, physics, and overall performance**.
  - Prevents lag during intense gameplay and ensures smooth multitasking.
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#### ⚡ 3. RAM (Memory)

- Upgrade to at least **8GB or 16GB RAM**.
  - Helps in **faster loading and smoother gameplay**.
  - Reduces lag when running the game and background apps.
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#### 4. Storage (SSD Upgrade)

- Switching from HDD to SSD makes **game loading much faster**.

- Improves overall system responsiveness.
  - FIFA will start quickly and load matches faster.
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## 🔌 5. Power Supply Unit (PSU)

- Needed if you upgrade GPU/CPU.
  - Ensures **stable and sufficient power supply** to all components.
  - Prevents system crashes or shutdowns.
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## ❄️ 6. Cooling System (Optional but Important)

- Keeps the system **cool during long gaming sessions**.
- Prevents overheating and performance drops.